

STUDENT-SUBMITTED QUESTIONS & ANSWERS

PART 2

EXAM III REVIEW DAY

ELEMENTARY LINEAR ALGEBRA (MATH 2250-03)

SPRING, 2019

QUESTIONS

1. I am struggling to understand the difference between eigenvalues, eigenvectors, and eigenspaces. Is it safe to assume that these are similar to how $\text{Col}(A)/\text{Nul}(A)$ /standard vectors work?

To answer your question, we'll describe the difference between eigenvalues, eigenvectors, and eigenspaces by giving the definitions of each of these things and highlighting the properties that vary between them. We'll start with eigenvalues and work our way up to eigenspaces.

Given an $n \times n$ matrix A , a scalar $\lambda \in \mathbb{R}$ is an *eigenvalue* of A if there exists a nonzero vector $x \in \mathbb{R}^n$ such that $Ax = \lambda x$.

So, eigenvalues are just some scalars (i.e. some numbers) associated to a matrix – but they're *particularly special* scalars. If this doesn't seem like a good explanation, then you might find it easier to think of eigenvalues like this instead:

Given an $n \times n$ matrix A , a scalar $\lambda \in \mathbb{R}$ is an *eigenvalue* of A if the homogeneous system $(A - \lambda I)x = 0$ has a nontrivial solution.

Using the invertible matrix theorem, we know that if $A - \lambda I$ has a nontrivial solution, then it isn't invertible. So, in light of this, an eigenvalue of an $n \times n$ matrix A is just a scalar $\lambda \in \mathbb{R}$ such that the matrix $A - \lambda I$ is not invertible.

The *eigenvectors* of an $n \times n$ matrix A are intimately tied to the *eigenvalues* of A . To be precise, we have the following definition:

Given an $n \times n$ matrix A , a nonzero vector $x \in \mathbb{R}^n$ is an *eigenvector* of A if there exists a scalar $\lambda \in \mathbb{R}$ such that $Ax = \lambda x$.

We can also define eigenvectors in terms of the matrix $A - \lambda I$ as we did with eigenvalues:

Given an $n \times n$ matrix A , a vector $x \in \mathbb{R}^n$ is an *eigenvector* of A if x is a nontrivial solution to the homogeneous system $(A - \lambda I)x = 0$ for some scalar $\lambda \in \mathbb{R}$.

As you can see, irrespective of the way we want decide to state these concepts, the definitions of eigenvectors and eigenvalues are virtually the same – the only difference is that we switched a part of the sentence around. Moreover, these definitions each *require each other*. That is, given an $n \times n$ matrix A , a vector $x \in \mathbb{R}^n$ can *only* be an eigenvector of A if there exists an eigenvalue $\lambda \in \mathbb{R}$ of A such that $Ax = \lambda x$, and a scalar $\lambda \in \mathbb{R}$ can *only* be an eigenvalue of A if there exists an eigenvector $x \in \mathbb{R}^n$ of A such that $Ax = \lambda x$.

This almost seems to say that eigenvalues and eigenvectors *come in pairs*, but there's a slight catch here that makes this not *quite* true. Namely, suppose A is an $n \times n$ matrix and $Ax = \lambda x$ for some nonzero vector $x \in \mathbb{R}^n$ and some scalar $\lambda \in \mathbb{R}$. Then, if $c \in \mathbb{R}$ is any scalar, we *also*

have that

$$\begin{aligned}A(cx) &= cAx \\ &= c\lambda x = \lambda(cx).\end{aligned}$$

So, if $x \in \mathbb{R}^n$ is an eigenvector of A , then any vector $y \in \text{span}\{x\}$ is *also* an eigenvector of A . Similarly, if $Ax = \lambda x$ and $Ay = \lambda y$, then for any scalars $c_1, c_2 \in \mathbb{R}$, we have

$$\begin{aligned}A(c_1x + c_2y) &= A(c_1x) + A(c_2y) \\ &= c_1Ax + c_2Ay \\ &= c_1\lambda x + c_2\lambda y = \lambda(c_1x + c_2y).\end{aligned}$$

So, if $x, y \in \mathbb{R}^n$ are both eigenvectors of A associated to the eigenvalue λ , then any vector $v \in \text{span}\{x, y\}$ is *also* an eigenvector of A associated to the eigenvalue λ .

To summarize the above, it's *not* quite true that eigenvectors and eigenvalues come in pairs, but it *is* true that eigenvalues and subspaces spanned by eigenvectors associated to those eigenvalues come in pairs. This motivates the define of an eigenspace, which can be most easily phrased as follows:

Given an $n \times n$ matrix A and an eigenvalue $\lambda \in \mathbb{R}$ of A , the *eigenspace* of A associated to the eigenvalue λ is the subspace $\text{Nul}(A - \lambda I)$ of $\mathbb{R}^n$.

More simply put, the eigenspace of A associated to an eigenvalue λ of A is just the set of all eigenvectors of A associated to λ along with the zero vector (which we throw in to get a genuine subspace of $\mathbb{R}^n$, instead of something clunkier and not as easy to work with).

So, to answer your question: Yes, eigenvectors work exactly like 'standard vectors' because they *are* just 'standard vectors' in the usual sense – the only difference is that they're *particularly special* vectors in relation to A . Similarly, yes, eigenspaces are exactly like null spaces because they *are* null spaces – they're just *particularly special* null spaces in relation to A . Eigenvalues are the things that tie both of these concepts together, but they're not too complicated either. Eigenvalues are simply scalars (i.e. numbers) – they're just *particularly special* scalars in relation to A .

To put it all in perspective, the whole theory of eigenvalues, eigenvectors, and eigenspaces is interesting, in part, because it gives a pretty deep connection between a lot of different objects: Matrices (A), scalars (eigenvalues of A), vectors (eigenvectors of A), and subspaces (eigenspaces of A). If you go further in mathematics, you'll find out that this is actually kind of amazing, and that following this line of thinking has allowed us to develop a large number of hugely powerful tools to address lots and lots of types of problems coming from a wide array of vastly different fields. We'll talk about this in the next question.

2. I do not really understand the point to eigenvalues. Calculating them makes sense I just do not see the big picture. Are there any real world applications for them? or do they just related to the theorems and formulas in linear algebra?

Absolutely, this stuff definitely has practical uses! In fact, the theory behind eigenvalues, eigenvectors, and eigenspaces has been used in a *truly huge* number of real-world applications which you probably use on a day-to-day basis in some form or another. Let's look at a few:

- (a) **Every Google search result you've ever seen:** Google's page rank algorithm ranks pages displayed in search results by using the largest eigenvector of the graph of the internet (click [this link](#) to read the Wikipedia page on Google's PageRank algorithm).
- (b) **Facial recognition:** Think, for example, of the face-lock feature on an iPhone. Things like this are done entirely by using a technique based directly in this theory, largely by using something called 'eigenfaces' (click [this link](#) to read the Wikipedia page on eigenfaces).

- (c) **Netflix movie recommendations:** Or, more generally 'low rank factorization for collaborative prediction', which is what Netflix does (or once did) to predict what rating you'll have for a movie you have not yet watched. It uses what's called a 'singular value decomposition' on a certain matrix A to compute a low-rank approximation to A , then uses this to throw away the smallest eigenvalues of the matrix $A^T A$ (click [this link](#) to read a paper on the topic).
- (d) **Image compression:** One of the main reasons why your phone can store as many relatively high-resolution pictures as it can and why we're now able to stream video as smoothly as we can is because we can compress image files by throwing away small eigenvalues of a certain matrix (click [this link](#) to read a paper on the topic and see some examples of image compression done in this fashion).
- (e) **Artificial intelligence:** Pretty much everything about neural networks and machine learning is linear-algebraic in nature to at least some degree, and almost always involves eigenvalues, eigenvectors, and eigenspaces in some way. Particularly, some crucial techniques, like dimension reduction and principle component analysis, work by projecting the largest eigenvalues of a certain matrix onto a certain subspace (click [this link](#) to read the Wikipedia page on principal component analysis).
- (f) **Segways, hoverboards, and self-driving cars:** Stabilizing things (like, for example, keeping a Segway or a hoverboard upright, or a self-driving car on the correct path) requires the use of techniques belonging to a field of mathematics called control theory. The Hautus lemma is a key result in control theory which let's us know when we can, among other things, stabilize things like segways and steer things like self-driving cars. The result comes pretty much entirely from some tricks involving the use of eigenvalues and eigenvectors (click [this link](#) to read the Wikipedia page on the Hautus lemma).

(Note: To properly cite my sources, some of these examples were taken or adapted from answers to [this question on math StackExchange](#).)

3. Could you explain how Linear transformations and eigenvectors interact with one another? For convenience, we'll keep everything finite-dimensional in this answer, but just as a side note, most of this answer is still true (with a variety of additional caveats) in infinite dimensions.

As we've learned, if V is a finite-dimensional vector spaces and T is a linear transformation $T : V \rightarrow V$, then there is a matrix for T with respect to any basis $\mathcal{B}$ for V . We can certainly compute the eigenvalues and eigenvectors of this matrix, so that's one way that eigenvalues/vectors and linear transformations interact. In particular, since every $n \times n$ matrix A defines a linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ by $T(x) = Ax$, the case where $\mathcal{B}$ is the standard basis of $\mathbb{R}^n$ covers our treatment of the topic in this class.

However, we can *also* define eigenvalues for linear transformations in a more general context:

Let V be a finite-dimensional real vector space and let $T : V \rightarrow V$ be a linear transformation.

We say that a scalar $\lambda \in \mathbb{R}$ is an *eigenvalue* of T if there exists a nonzero vector $v \in V$ such that $T(v) = \lambda v$. Similarly, we say that a non-zero vector $v \in V$ is an *eigenvector* of T if there exists a scalar $\lambda \in \mathbb{R}$ such that $T(v) = \lambda v$.

So, eigenvalues and eigenvectors can be defined in the very general context of arbitrary linear transformations between finite-dimensional real vector spaces in pretty much the exact same way as they're defined when we're dealing with $\mathbb{R}^n$ and an $n \times n$ matrix. If you're interested, this is useful in the infinite-dimensional analog I mentioned earlier because, among other reasons, it helps solve differential equations (click [this link](#) to read the Wikipedia page on eigenvalues and eigenfunctions for differential operators or [this link](#) to read the Wikipedia page on eigenfunctions in general).

4. I'm still struggling to understand the purpose of diagonalizing a matrix to get the eigenvalues and eigenvectors. We can get the eigenvalues through other means, and it's never mentioned again in the other sections. Is there some way where we would need to diagonalize a matrix in order to get the eigenvectors, because there's no other way? Or is the entire theorem just a shortcut that saves us time, and won't likely be used in any other way?

I think your confusion here comes from the fact that you've got this process slightly backwards: We don't use a diagonalization to find the eigenvalues and eigenvectors of a matrix, we use the eigenvalues and eigenvectors of a matrix to *find a diagonalization*.

This is a good question, and a lot of the confusion folks have about this is probably due to the types of problems teachers use in linear algebra classes. That is, while your teacher might hand you a diagonalization of a matrix A and ask you to find the eigenvalues and eigenvectors of A in homework problem, this is not really a situation that ever arises in practice. Teachers just use problems like this because it's an easy way to test students' understanding of some of the concepts involved in this topic, not because it's a realistic situation that commonly occurs.

On the other hand, however, computing eigenvalues and vectors in order to diagonalize a matrix *is* a very common real-world situation. The reason why it's common is because the types of matrices involved in a diagonalization (i.e. invertible matrices and diagonal matrices) are *very easy to work with* compared to most other matrices. That is, after you know the diagonalization $A = PDP^{-1}$ of a matrix A , you can compute A^n simply by taking the n^{th} power of the entries on the diagonal of D , which involves *way, way fewer* computations than simply multiplying A by itself n times.

Aside from the lower computational costs associated to calculations involving the diagonalization of a matrix and other practical implementation reasons, diagonalization is also important from a theoretical angle as well. Namely, diagonalization makes it easy for us to *understand* what certain linear transformations are doing to vectors. That is, if A is a diagonalizable matrix and $A = PDP^{-1}$, then when A is multiplied by a vector x (i.e. when we're computing Ax), all that happens is that:

- A first takes x and changes its coordinates from the standard basis $\mathcal{E}$ of $\mathbb{R}^n$ to a basis $\mathcal{B}$ of $\mathbb{R}^n$ formed by the columns of P (so, to a basis formed from the eigenvectors of A).
That is, $P^{-1}x = P^{-1}[x]_{\mathcal{E}} = [x]_{\mathcal{B}}$.
- A then scales each entry of the corresponding $\mathcal{B}$ -coordinate vector $[x]_{\mathcal{B}}$ by the corresponding entries on the diagonal of D (so, the eigenvalues of A).
That is, $DP^{-1}x = D[x]_{\mathcal{B}}$
- A finishes its transformation of x by undoing the change of coordinates and returning the resulting vector to the standard basis $\mathcal{E}$ of $\mathbb{R}^n$.
That is, $Ax = PDP^{-1}x = [D[x]_{\mathcal{B}}]_{\mathcal{E}}$.

5. My question is, can you go over an example of finding eigenspaces? Specially when the matrix is not in PDP^{-1} form.

Given an eigenvalue λ of an $n \times n$ matrix A , all we need to do to find the eigenspace associated to λ is compute $\text{Nul}(A - \lambda I)$. So, for example, $\lambda = 1$ is an eigenvalue of the matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. So, to find the eigenspace of A associated to the eigenvalue $\lambda = 1$, all we need to do is compute $\text{Nul}(A - I)$. Since

$$A - I = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

we can find $\text{Nul}(A - I)$ by solving the homogenous linear system $(A - I)x = 0$, i.e.

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

As a system of equations, this is just

$$\begin{aligned}x_1 + x_2 &= 0 \\x_1 + x_2 &= 0.\end{aligned}$$

So, our solution has x_1 free and $x_2 = -x_1$ (or, if you prefer, x_2 free and $x_1 = -x_2$). In parametric form, this means that our solutions look like

$$x = \begin{bmatrix} x_1 \\ -x_1 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

As such, the eigenspace of A associated to the eigenvalue $\lambda = 1$ is just the subspace

$$\text{span} \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}.$$

6. Will we be asked to find a D and P from a matrix A on our own? If so, could you walk through an example or at least specify the steps you take to find those matrices? The examples from class and the homework aren't clear on the process to do so.

From the Diagonalization Theorem, finding a diagonal matrix D and an invertible matrix P for a diagonalizable matrix A can be done by computing the eigenvalues and eigenvectors of A .

For example, let's take the matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ and find a diagonal matrix D and an invertible matrix P such that $A = PDP^{-1}$.

To start, let's begin by finding the eigenvalues of A . To do so, we just need to solve the equation $\det(A - \lambda I) = 0$ for λ . So, computing, we have

$$\begin{aligned}\det(A - \lambda I) &= 0 \\ \det \left(\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) &= 0 \\ \det \left(\begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix} \right) &= 0 \\ (2-\lambda)^2 - (1)(1) &= 0 \\ \lambda^2 - 4\lambda + 3 &= 0 \\ (\lambda - 1)(\lambda - 3) &= 0.\end{aligned}$$

So, it's easy to see that the eigenvalues of A are just $\lambda_1 = 1$ and $\lambda_2 = 3$. In fact, since an $n \times n$ matrix with n distinct eigenvalues is diagonalizable (this is Theorem 6 in section 5.3 of your textbook), this tells us that A is diagonalizable (since A is a 2×2 matrix with 2 distinct eigenvalues). So, we shouldn't run into any troubles finding our desired matrices P and D .

Now, to continue, let's find the corresponding eigenvectors of A . To do so, we need to solve the systems $Av_1 = \lambda_1 v_1$ for v_1 and $Av_2 = \lambda_2 v_2$ for v_2 . Since these can be re-written as $(A - \lambda_1 I)v_1 = 0$ and $(A - \lambda_2 I)v_2 = 0$, we just need to find a vector v_1 in the null space of $A - \lambda_1 I$ and a vector v_2 in the null space of $A - \lambda_2 I$. Since

$$A - \lambda_1 I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad A - \lambda_2 I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

we can take v_1 to be any solution to the system

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

and take v_2 to be any solution to the system

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

In the first case, we've already computed an answer to this in question 5, so we can take $v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. In the second case, we get $x_1 = x_2$ and can take $v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

So, by using the Diagonalization Theorem, we have

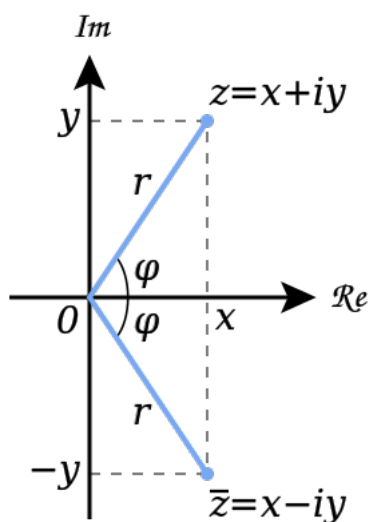
$$\begin{aligned} A &= P D P^{-1} \\ \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} &= \begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} v_1 & v_2 \end{bmatrix}^{-1} \\ \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} &= \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}^{-1}. \end{aligned}$$

That is, $P = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$ and $D = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$.

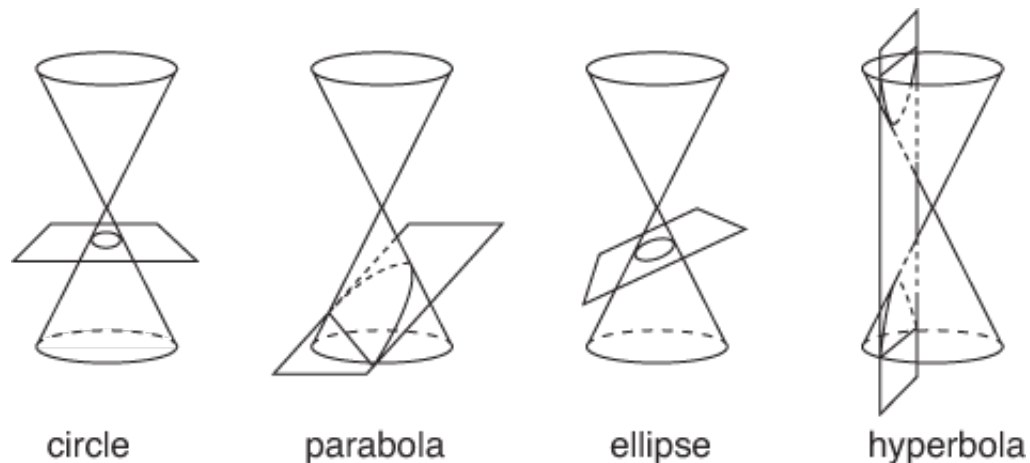
7. I was wondering, what exactly is the complex conjugate used for? I understand what they are in terms of like $z = 4 + 3i$ and the conjugate would be $\bar{z} = 4 - 3i$, but how is this useful in solving problems? What do the conjugates do?

The operation of complex conjugation is important for a variety of reasons (most of which are too technical or unrelated to get into here), but for the purposes of this class, complex conjugates come up in linear algebra because complex roots to polynomials with real coefficients come in *conjugate pairs*. That is, the polynomial $x^2 + 1$ has a root at $x = i$ and $\bar{x} = -i$. So, if A is a real $n \times n$ matrix, then when we're solving $\det(A - \lambda I) = 0$ for λ , we're solving for the zeros of a polynomial with real coefficients. Therefore, if this polynomial has any complex roots, those complex roots will come in conjugate pairs. Using the notation of complex conjugation just makes them easier to talk about.

Now, to understand, on an intuitive level, *why* the complex roots of polynomials with real coefficients come in conjugate pairs like they do, let's think of a complex number z as a vector in $\mathbb{R}^2$. If we do this, then notice that $\operatorname{Re}(z)$ and $\operatorname{Im}(z)$ are *projections*. That is, $\operatorname{Re}(z)$ is the projection of the complex number z onto the real axis and so on. In this case, the complex conjugate of z is the *only other complex number* such that $\|z\| = \|\bar{z}\|$ and $\operatorname{Re}(z) = \operatorname{Re}(\bar{z})$. That is, they have the same projection onto the real line and the same magnitude (or length, if you prefer). The picture below shows this quite clearly:



So, the fact that complex roots to polynomials with real coefficients come in conjugate pairs says that such functions are zero at points in $\mathbb{C}$ which are *symmetric* about the real axis. Why is this? Well, consider the fact that we can always factor a polynomial with real coefficients over the reals into some number of linear factors of the form $(ax + b)$ (corresponding to real roots) and some number of quadratic factors $(ax^2 + bx + c)$ (corresponding to complex roots). With this in mind, the symmetry in the complex roots of a polynomial with real coefficients can be seen as a consequence of the fact that the graph of a quadratic function is a parabola, which is a *conic section* (click [this link](#) to read the Wikipedia page about conic sections), as in the picture below:



Put loosely, the symmetry of this conic induces the symmetry of the complex roots of a polynomial with real coefficients about the real axis. Hence, the complex roots come in conjugate pairs. If you'd like a further explanation of complex conjugation, click [this link](#) to read some notes on the topic by Paul Dawkins, an excellent professor at Lamar University who has some truly awesome online math notes.

8. In regards to Homework 9, Problem 4:

We know that we seek the complex eigenvalues of the 2×2 matrix A by letting A act on $\mathbb{C}^2$. Does letting A , an $n \times n$ matrix act on $\mathbb{C}^n$ mean that there's any special operation attached to how you should go about solving the problem (finding the eigenvalues/eigenspace) or is it merely saying that we are allowed to use complex numbers?

i.e. If we set up a problem on the final and we have to solve for a complex eigenvalue, do you have to state something like "Now, because there aren't any 'real' eigenvalues, we must let our matrix A act on $\mathbb{C}^n$ to allow for the use of complex eigenvalues."

Great question! The language of letting a matrix "act" on $\mathbb{C}^n$ is actually pretty precise and technical, in the sense that it's an example of what's known as a 'group action' (click [this link](#) to read the Wikipedia page about group actions). But, while there is this underlying formalism that makes all of this work on a deep level, it's pretty much irrelevant to anything we're doing in this class. As it turns out, your answer is exactly correct: When we say "Let A act on $\mathbb{C}^n$ " it really just means that we're allowed to use complex numbers now.

In linear algebra language, the reason for this is pretty much just due to the definition of an eigenvalue or an eigenvector: By definition, an eigenvector must live in the space on which the matrix acts, and the eigenvalue must belong to the underlying field of that space. So, if A is a real 2×2 matrix acting on $\mathbb{R}^2$ with characteristic polynomial $\lambda^2 + 1$, it is correct to say that A doesn't have any eigenvalues. Because this makes things trickier, we instead decide to just *treat A like it's a complex 2×2 matrix* (which works because real numbers are *also* complex numbers—they just have zero imaginary part) acting on $\mathbb{C}^2$. So, we just treat our matrix with entries in our original underlying field ($\mathbb{R}$, in this case) as if it were a matrix with entries belonging a 'larger' field ($\mathbb{C}$, in this case) in which the characteristic polynomial has roots. This is an example of something called a 'field extension' (click [this link](#) to read the Wikipedia page about field extensions), but that isn't particularly important for this class. As far as we need to be concerned at this point, your intuition is exactly correct.

9. I am having a hard time understanding how to solve complex eigenvalues. In problem 5 on homework 9 I do not know where the i 's come from in the second part of the solution. I believe it must have something to do with using the characteristic equation upon this problem, but I do not understand how solving what is produced by that yields an equation with complex numbers. I feel like I understand a few things about complex numbers, such as the fact that they rotate things, but I feel like they are still pretty weird especially when I am working with them in problems. If you could please shed a little more light on this during the review that would be fantastic.

Complex numbers aren't actually too bad when you get down to it. In a very precise sense, they're just the real numbers, but with a special "extra" number added (which Descartes decided to call i back in 1637) that satisfies $i^2 = -1$. Amazingly, while polynomials with real coefficients don't always have real roots, by simply adding this single extra number, every polynomial with complex coefficients has complex roots. This is really special in mathematics, for a lot of reasons I won't get into here. But as far as Problem 5 on Homework 9 goes, it works in exactly the same way as question 5 in this document – the only difference is that you're now solving these equations and systems using *complex* numbers instead of only *real* numbers. So, from a linear algebra standpoint, the process is exactly the same, the algebra is just over a different field. Unfortunately, finding the complex roots of polynomials and solving polynomial equations involving complex numbers is more of a standard algebra topic than a linear algebra topic, and would be a bit outside the scope of this document to explain. But, if you want a review of complex numbers and how to do algebra with them, I highly recommend clicking either [this link](#) or [this link](#) to read some notes and refresh your memory on the topic!

10. There was a theorem in Section 5.5 that said that A , a real 2×2 matrix with non-real eigenvalues, could be factored as $A = PCP^{-1}$. In this, P was described as the matrix whose columns are the real and imaginary components of the eigenvectors, and C was $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$. What is the purpose of this factorization? With $\begin{bmatrix} b & a \end{bmatrix}$ diagonalization, the factored form gave us a clear way to see the eigenvalues and eigenvectors of the factored matrix, but what is the purpose of this factorization? Additionally, can we always be certain that P is invertible? What happens for a matrix larger than 2×2 where there may be more than one a and b values?

This is a fun question! I believe you're referring to the following theorem:

Let A be a real 2×2 matrix with a complex eigenvalue $\lambda = a - bi$ and an associated eigenvector $v \in \mathbb{C}^2$. Then, $A = PCP^{-1}$, where $P = \begin{bmatrix} \text{Re}(v) & \text{Im}(v) \end{bmatrix}$ and $C = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$.

To answer part of your question, we can tell that P is invertible because $\text{Re}(v)$ and $\text{Im}(v)$ are linearly independent (click either [this link](#) or [this link](#) to see some math StackExchange questions and answers which explain why), and any square matrix with linearly independent columns is invertible. The purpose of this factorization is pretty much the same as the purpose of diagonalization (which I'll skip talking about again, since it's addressed in the answer to Question 4 in this document).

Finally, to answer your question about what happens in higher dimensions: In short, things get more complicated. In general, if an $n \times n$ matrix A doesn't have n linearly independent eigenvectors in $\mathbb{R}^n$, we can't diagonalize A . In this case, A is known as a 'defective' matrix, and the best we can do is factor A using what's known as a Jordan normal form of A , which can be viewed loosely as a generalization of diagonalization. To find out more about this, click [this link](#) to read the portion of the Wikipedia article on the Jordan normal form regarding real matrices.

11. A non-linear algebra related question, what exactly is fidlar? I've noticed two shirts and the tattoo. Is it a band? A cult? Just curious really.

I thought it was obvious, but FIDLAR is an acronym for "Forget It Dad, Linear Algebra Rocks." Some scoundrels have told me that it might also mean (pardon my French) "Fuck it dog, life's a risk" and that there might be a [solid punk band](#) with the same name, but I wouldn't know anything about that.